

Nuclear Physics Primer for Nuclear Astrophysics

Note for NS 3035

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“A ship in harbor is safe, but that is not what ships are built for” — John A. Shedd

*In the first three classes we discussed the foundations of a simple stellar model. We began with non rotating uniform stars and developed several important concepts such as the free fall timescale, hydrostatic equilibrium, and the virial theorem (I encourage you to practice these derivations on your own). Using kinetic theory, we realized that most ordinary stars exist in a **non relativistic** regime where the thermal motion of particles dominates the internal physics.¹ In Classes 3 and 4, we stepped back to look at the bigger astrophysical picture. Our prime question was how stars are formed ? We discussed the evolution from the interstellar medium (ISM) to giant molecular clouds (GMCs), the formation of open clusters and the importance of the Jeans criterion in initiating gravitational collapse. We did not want to remain at a surface level discussion by simply looking at pictures and memorizing a few fancy names. Those kinds of courses are for toddlers!. As science students, our goal is to develop deeper physical intuition even when the concepts initially feel uncomfortable.*

From this week onward, we will pause our astronomy discussions and move toward the more familiar territory of nuclear science. I understand that the first few classes may have felt somewhat overwhelming, but I believe you will become much more comfortable from this point onward. However, even if some of the coming weeks feel like Greek to you, I invite you to reflect on the quote placed at the beginning of this document, The best learning often begins just beyond our comfort zone. It means that, truly understanding this subject is rarely comfortable at the beginning. Derivations may feel messy, and sometimes the mathematics may appear intimidating. But that discomfort is often a sign that real learning is taking place. So as we move deeper into nuclear astrophysics hang on tight and prepare for rough seas ahead. We are leaving the calm harbor and sailing into open waters !!

¹We also briefly mentioned the ultra relativistic regime where electron degeneracy pressure becomes important

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Chapter 1

Nomenclature

Before studying thermonuclear reactions inside stars, we must first establish the basic language and notation of nuclear physics. The **nucleus** is the dense central region of an atom containing protons and neutrons and therefore almost all of the atom's mass. The plural form of nucleus is **nuclei**

$$\text{one nucleus} \longrightarrow \text{many nuclei}$$

The particles contained inside the nucleus are called **nucleons**. Nucleons consist of two types of particles

$$\text{nucleons} = \text{protons} + \text{neutrons}$$

In simple terms

- The **nucleus** is the bound object itself
- **Nucleons** are the particles inside the nucleus
- **Nuclei** is simply the plural form of nucleus

Protons carry positive electric charge, while neutrons are electrically neutral. The number of protons inside a nucleus is called the **atomic number** and is denoted by Z . The total number of nucleons inside the nucleus is called the **mass number** and is denoted by A .

A nucleus is therefore written in the form

A_ZX

where X is the chemical symbol of the element. The number of neutrons is

$$N = A - Z$$

For example, carbon-12 is written as

$${}^{12}_6\text{C}$$

which means that the nucleus contains six protons and six neutrons.

Nuclei that possess the same number of protons but different numbers of neutrons are called **isotopes**. Hydrogen provides a simple example

$${}^1\text{H}, \quad {}^2\text{H}, \quad {}^3\text{H}$$

Even though all of these nuclei correspond to hydrogen, they possess different physical properties because their neutron numbers are different.

1.1 Fundamental Forces of Nature

Inside the nucleus, positively charged protons strongly repel one another through the Coulomb force. If electromagnetism were the only interaction present, stable nuclei could not exist. Therefore, another force must exist to keep nuclei bound together.

To understand this, we briefly introduce the four fundamental forces of nature

1. Gravitational force
2. Electromagnetic force
3. Strong nuclear force
4. Weak nuclear force

Gravity dominates the large scale structure of the Universe and governs the dynamics of stars, galaxies, giant molecular clouds (GMCs), and the interstellar medium (ISM), which we briefly discussed in the first few classes. However, at nuclear scales gravity is extremely weak and can safely be neglected.

The electromagnetic force acts between electrically charged particles. Inside the nucleus, positively charged protons repel each other through Coulomb repulsion. Stable nuclei therefore cannot exist through electromagnetism alone.

The **strong nuclear force** is responsible for overcoming this repulsion and binding protons and neutrons together. Unlike gravity or electromagnetism, the strong interaction acts only over extremely short distances, roughly of the order

$$\sim 10^{-15} \text{ m}$$

corresponding to typical nuclear sizes. At these very small scales, however, the strong interaction becomes enormously powerful and dominates over electromagnetic repulsion.

The **weak nuclear force** is responsible for processes such as beta decay and neutrino interactions. Although much weaker than the strong interaction, it plays a fundamental role in stellar nucleosynthesis and thermonuclear reactions inside stars. In fact, the proton–proton reaction powering the Sun proceeds through the weak interaction. This is one of the major reasons why the Sun burns slowly and remains stable over billions of years.

1.2 Basics

The energy released in nuclear reactions originates from the difference between the masses of the initial nuclei and the final nuclei. This difference is called the **mass defect**. For a nucleus containing Z protons and N neutrons, the mass defect is written as

$$\Delta M = Zm_p + Nm_n - M_{\text{nucleus}}$$

where

- M_{nucleus} is the mass of the nucleus,
- m_p is the proton mass,
- m_n is the neutron mass,
- Z is the atomic number,
- N is the neutron number.

A bound nucleus always has a slightly smaller mass than the total mass of its free protons and neutrons. The “missing” mass appears as binding energy.

The nuclear binding energy is therefore

$$B = \Delta Mc^2$$

where c is the speed of light. The binding energy represents the energy required to completely separate the nucleus into individual free nucleons.

In nuclear astrophysics, fusion reactions involving light nuclei are energetically favorable because fusion generally increases the binding energy per nucleon. This is one of the fundamental reasons why stars are able to generate energy through thermonuclear fusion.

An important laboratory example of controlled thermonuclear fusion research is ITER¹ <https://www.iter.org/>. ITER, located in France, is one of the largest magnetic confinement plasma physics experiments in the world. Its goal is to reproduce fusion conditions similar to those found inside stars.

Sun : natural fusion laboratory

ITER : artificial fusion laboratory

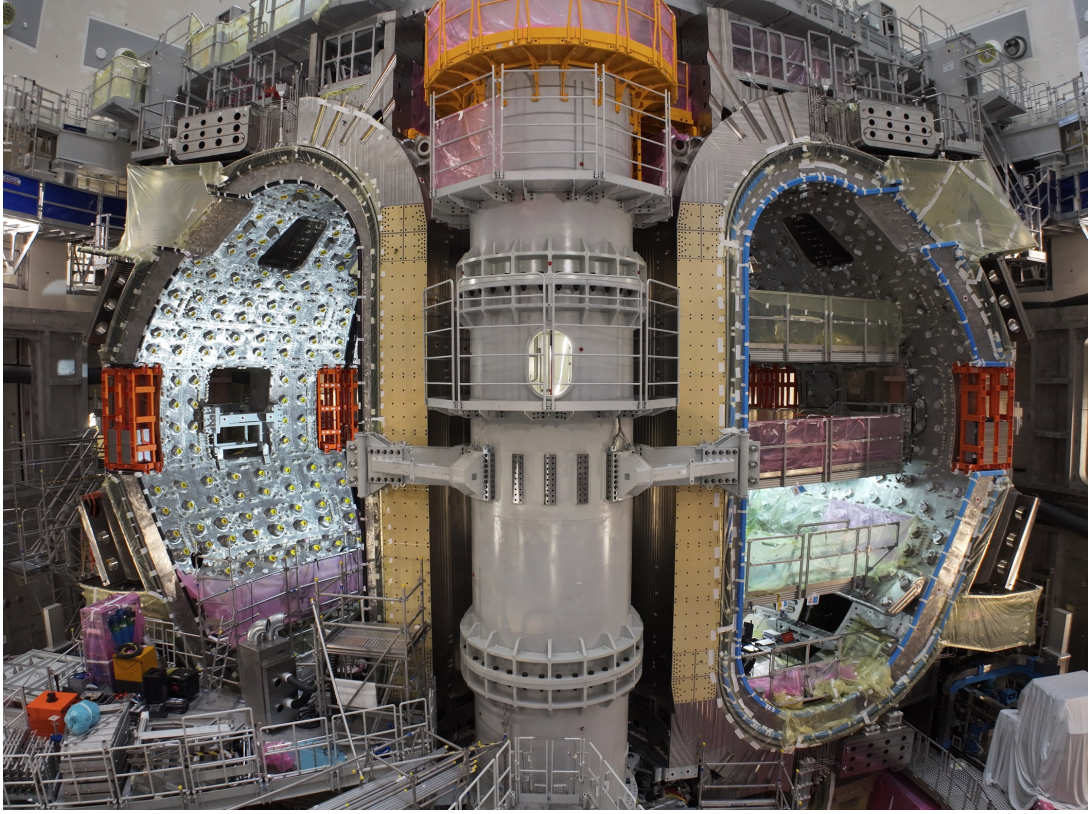


Figure 1.1: A cross-section look at the assembly of the ITER tokamak reactor, highlighting the towering central solenoid column and the D-shaped vacuum vessel chamber designed to magnetically confine a thermonuclear burning plasma. Inside the ITER assembly facility, engineering a human-controlled environment where light nuclei can conquer the Coulomb barrier to achieve self-sustaining thermonuclear fusion. For project details, see the official website: <https://www.iter.org/>

1.2.1 General Nuclear Reaction

A general nuclear reaction may be written as

$$1 + 2 \longrightarrow 3 + 4$$

where

- particles 1 and 2 are the initial reactants,
- particles 3 and 4 are the final reaction products.

Nuclear Q-value

One of the most important quantities in nuclear astrophysics is the reaction **Q-value**, which measures the energy released or absorbed during a nuclear reaction.

Using nuclear masses, the nuclear Q-value is

$$Q_n = [M_{n,1} + M_{n,2} - (M_{n,3} + M_{n,4})] c^2$$

¹ITER = International Thermonuclear Experimental Reactor

Physically,

$$Q > 0 \quad \Rightarrow \quad \text{energy is released}$$

$$Q < 0 \quad \Rightarrow \quad \text{energy is absorbed}$$

Exothermic reactions ($Q > 0$) are especially important in stellar fusion because they provide the energy responsible for stellar luminosity.

Atomic Q-value

In many practical calculations, atomic masses are used instead of nuclear masses because atomic mass tables are experimentally easier to obtain. Using atomic masses, the Q-value becomes

$$Q_a = [M_{a,1} + M_{a,2} - (M_{a,3} + M_{a,4})] c^2$$

The atomic and nuclear Q-values are related through electron binding energy corrections

$$Q_a = Q_n + \Delta B_e$$

where ΔB_e represents the difference in total electron binding energies between the initial and final atoms.

For most nuclear astrophysics calculations, these corrections are extremely small and may usually be neglected

$$Q_a \approx Q_n$$

1.2.2 Exceptions to the Approximation $\Delta B_e \approx 0$

Although the approximation

$$\Delta B_e \approx 0$$

is valid for many reactions, there are important exceptions. Reactions involving electrons or positrons require special attention because electron masses and electron binding energies contribute explicitly to the reaction energetics.

In such cases, the atomic mass is related to the nuclear mass through

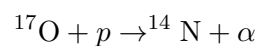
$$M_a = M_n + Zm_e - \frac{B_e(Z)}{c^2}$$

where

- M_a is the atomic mass,
- M_n is the nuclear mass,
- m_e is the electron mass,
- $B_e(Z)$ is the total electron binding energy.

1.2.3 Example : Reaction Q-value

Using Nuclear Masses



Using nuclear masses, the Q-value is

$$Q_n = [M_n(^{17}\text{O}) + M_n(p) - M_n(^{14}\text{N}) - M_n(\alpha)] c^2$$

Alternatively, using mass excess values,

$$Q_n = (ME)_{^{17}\text{O}} + (ME)_p - (ME)_{^{14}\text{N}} - (ME)_\alpha$$

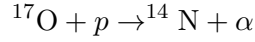
Substituting the experimental values gives

$$Q_n \approx 1.19 \text{ MeV}$$

Since $Q > 0$, the reaction releases energy.

Using Atomic Masses

The same reaction may also be calculated directly using atomic masses



$$Q_a = [M_a(^{17}\text{O}) + M_a(p) - M_a(^{14}\text{N}) - M_a(^4\text{He})] c^2$$

Substituting tabulated atomic masses,

$$Q_a = [16.999131 + 1.007825 - 14.003074 - 4.002603] \times 931.5$$

$$Q_a \approx 1.19 \text{ MeV}$$

The result agrees closely with the nuclear Q-value because electron binding energy corrections are extremely small.

Including Electron Binding Energy Corrections

The relation between atomic and nuclear Q-values is

$$Q_a = Q_n + \Delta B_e$$

where

$$\Delta B_e = B_e(Z_1) + B_e(Z_2) - B_e(Z_3) - B_e(Z_4)$$

Suppose

$$Q_n = 1.190 \text{ MeV}$$

and

$$\Delta B_e = -0.003 \text{ MeV}$$

Then

$$Q_a = 1.190 - 0.003$$

$$Q_a = 1.187 \text{ MeV}$$

Since the correction is extremely small, most stellar reaction calculations use the approximation

$$Q_a \approx Q_n$$

Example : Proton–Proton Reaction Inside the Sun

$$p + p \rightarrow d + e^+ + \nu_e$$

Using atomic hydrogen masses,

$$Q = (2M_H - M_d)c^2$$

which gives

$$Q \approx 1.44 \text{ MeV}$$

Using nuclear masses only,

$$Q = (2M_p - M_d)c^2$$

$$Q \approx 0.93 \text{ MeV}$$

The difference originates from electron masses included in the atomic hydrogen masses.

Chapter 2

Nuclear Structure and Stability

In the previous chapter we introduced the basic grammar of nuclear physics, including nuclei, nucleons, isotopes, binding energy, and nuclear reaction energetics. We now move one step deeper and investigate how nuclei are structured and why some nuclei are stable while others are unstable. Understanding nuclear structure is one of the central foundations of nuclear astrophysics because stellar nucleosynthesis, radioactive decay, neutron capture processes, supernova explosions and compact object formation all depend strongly on nuclear stability. At first glance, the nucleus appears to be a tiny collection of protons and neutrons confined inside an extremely small region of space. However, nuclei are not random collections of particles. They possess highly organized quantum structures governed by the strong interaction, quantum mechanics.

2.1 Characteristic Size of Nuclei

One of the first important questions in nuclear physics is

How large is a nucleus?

Experimental scattering experiments show that nuclear radii approximately follow the relation

$$R = R_0 A^{1/3}$$

where

- R is the nuclear radius,
- $R_0 \approx 1.2 \times 10^{-15}$ m,
- A is the mass number.

Since

$$1 \text{ fm} = 10^{-15} \text{ m}$$

nuclear sizes are typically of the order of a few femtometers. For example,

$$R(^{56}\text{Fe}) = 1.2 \times 56^{1/3} \text{ fm} \approx 4.6 \text{ fm}$$

This relation reveals an important physical property of nuclei. Since the radius scales as $A^{1/3}$, the nuclear volume scales approximately as

$$V \propto R^3 \propto A$$

which means nuclear density remains approximately constant for almost all stable nuclei

2.2 Nuclear Density

The density of nuclear matter is extraordinarily large compared to ordinary matter. Typical nuclear densities are approximately

$$\rho_{\text{nuclear}} \sim 10^{17} \text{ kg m}^{-3}$$

To appreciate how extreme this density is, consider the following:

- A teaspoon of neutron star matter would weigh billions of tons
- Nuclear matter is far denser than ordinary solids or liquids
- Such enormous densities naturally appear inside neutron stars

The near constant nuclear density strongly suggests that nucleons interact mainly with nearby particles rather than all particles inside the nucleus simultaneously. This property is known as **saturation** of the strong nuclear force

2.3 Binding Energy Per Nucleon

In the previous chapter we introduced the total nuclear binding energy

$$B = \Delta Mc^2$$

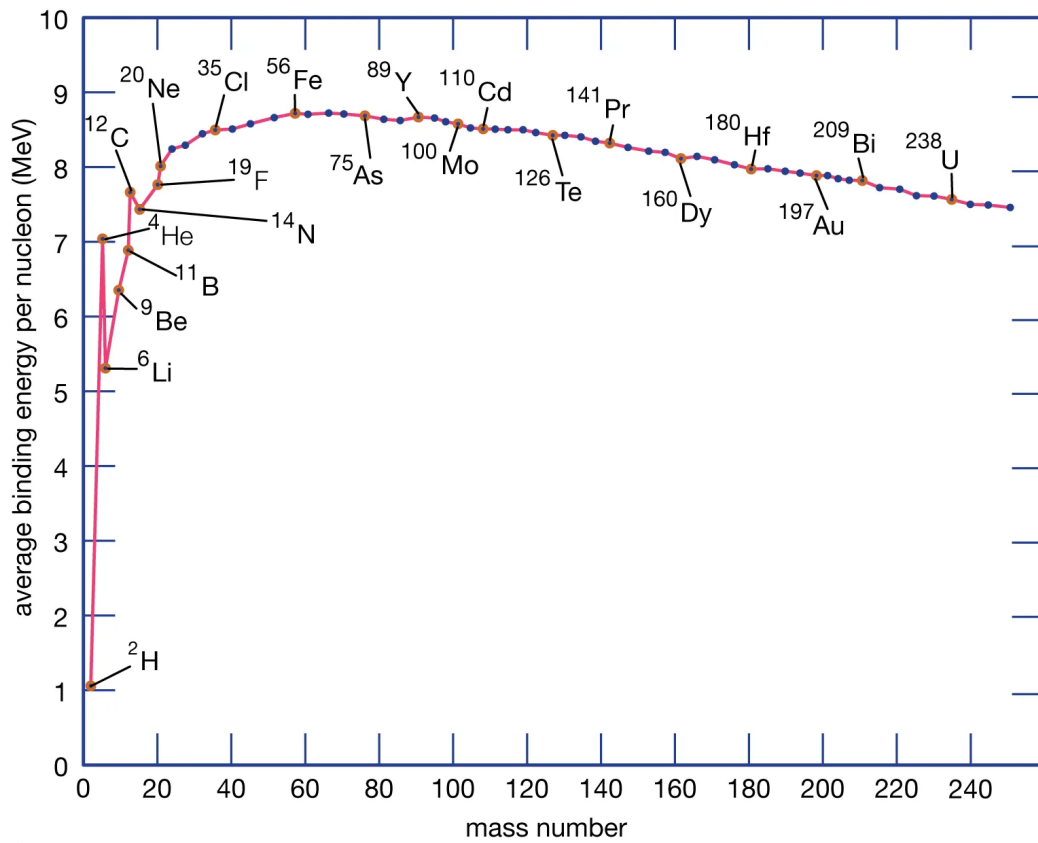
However, a more physically useful quantity is the *binding energy per nucleon*

$$\frac{B}{A}$$

which measures the average binding energy contributed by each nucleon. The binding energy per nucleon provides a direct measure of nuclear stability.

- Larger B/A generally corresponds to greater stability
- Smaller B/A corresponds to less stable nuclei

One of the most important results in nuclear physics is that the binding energy per nucleon is not constant across all nuclei. Light nuclei possess relatively small binding energies per nucleon, while intermediate mass nuclei become increasingly stable. The binding energy curve reaches a maximum near iron and nickel nuclei, particularly around ^{56}Fe and ^{62}Ni . After this region, the binding energy per nucleon slowly decreases for very heavy nuclei.



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Figure 2.1: Nuclear binding energies per nucleon, shown as a function of atomic mass number (A). The curve illustrates the region of enhanced stability near the iron peak (^{56}Fe), which dictates why lighter elements release energy via nuclear fusion while heavier elements undergo nuclear fission <https://www.britannica.com/science/nuclear-binding-energy>

2.4 Why Fusion Releases Energy

The binding energy curve immediately explains why thermonuclear fusion powers ordinary stars. Light nuclei such as hydrogen and helium possess relatively low binding energy per nucleon. When these nuclei fuse together, the resulting heavier nucleus usually possesses a larger binding energy per nucleon. Therefore,

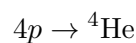
final nucleus more tightly bound

which means mass is converted into energy according to

$$E = mc^2$$

This released energy becomes the source of stellar luminosity.

For example,



converts a small fraction of mass into energy inside the Sun.

This process powers main sequence stars over billions of years.

2.5 Why Fission Releases Energy

The same binding energy curve also explains why nuclear fission can release energy.

Very heavy nuclei such as uranium possess smaller binding energies per nucleon compared to intermediate mass nuclei. If a heavy nucleus splits into smaller fragments closer to the iron region, the final nuclei become more tightly bound.

Therefore,

heavy nucleus $\rightarrow$ more stable fragments

and energy is released.

Thus:

- Fusion releases energy for light nuclei,
- Fission releases energy for heavy nuclei.

2.6 Valley of Stability

Not all nuclei are stable. Some nuclei spontaneously decay into other nuclei through radioactive processes.

If we plot neutron number N against proton number Z , stable nuclei form a region known as the

Valley of Stability

Light stable nuclei generally satisfy

$$N \approx Z$$

because equal numbers of protons and neutrons minimize the nuclear energy.

However, for heavier nuclei additional neutrons become necessary to help stabilize the nucleus against increasing Coulomb repulsion between protons.

Therefore heavy stable nuclei satisfy

$$N > Z$$

Nuclei far away from the valley of stability are unstable and undergo radioactive decay.

2.7 Radioactive Instability

Unstable nuclei attempt to move toward more stable configurations through radioactive decay processes such as

- alpha decay,
- beta decay,
- gamma decay,
- electron capture.

Radioactive decay plays a central role in nuclear astrophysics because

- unstable nuclei are continuously produced inside stars,
- radioactive decay powers supernova light curves,
- neutron rich nuclei dominate r-process nucleosynthesis,
- decay chains influence elemental abundances throughout the Universe.

2.8 Magic Numbers

Experimental observations show that certain nuclei possess unusually high stability. These nuclei correspond to specific numbers of protons or neutrons known as

2, 8, 20, 28, 50, 82, 126

called the **magic numbers**.

Nuclei with magic numbers often possess

- enhanced stability,
- larger binding energies,
- smaller reaction probabilities,
- longer lifetimes.

The existence of magic numbers strongly suggests that nucleons occupy discrete quantum energy levels inside the nucleus, similar to electrons inside atoms.

This idea eventually leads to the nuclear shell model.

2.9 Neutron Rich and Proton Rich Nuclei

In astrophysical environments, many nuclei exist far from stable conditions.

Neutron Rich Nuclei

Neutron rich nuclei contain far more neutrons than stable nuclei.

These nuclei become extremely important in

- r-process nucleosynthesis,
- neutron star mergers,
- supernova explosions.

Proton Rich Nuclei

Proton rich nuclei contain excess protons and often undergo

- beta decay,
- proton emission,
- electron capture.

Such nuclei become important in explosive stellar environments such as

- X-ray bursts,
- novae,
- rp-process nucleosynthesis.

2.10 Drip Lines

There exists a limit to how many neutrons or protons a nucleus can hold.

If too many neutrons are added, the nucleus reaches the

neutron drip line

beyond which neutrons immediately escape from the nucleus.

Similarly, excessive proton numbers produce the

proton drip line

beyond which protons are no longer bound.

Many nuclei involved in astrophysical nucleosynthesis lie extremely close to these drip lines.

2.11 Nuclear Stability and Stellar Evolution

Nuclear stability directly controls the life cycles of stars.

- Stable fusion reactions power stars.
- Radioactive decay modifies elemental abundances.
- Neutron capture creates heavy nuclei.
- Nuclear instability drives explosive nucleosynthesis.
- Iron group nuclei eventually terminate ordinary stellar fusion.

In fact, the entire chemical evolution of the Universe is ultimately controlled by nuclear structure and stability.

Without nuclear binding, quantum shell structure, and radioactive decay, stars, planets, and biological elements could never exist.

2.12 Summary

In this chapter we introduced the basic physical properties governing nuclear structure and stability.

The most important ideas are

- nuclear radii scale approximately as $A^{1/3}$,
- nuclei possess nearly constant density,
- binding energy per nucleon determines nuclear stability,
- fusion powers stars because light nuclei move toward greater stability,

- heavy nuclei may release energy through fission,
- stable nuclei form a valley of stability,
- magic numbers arise from quantum shell structure,
- unstable nuclei play major roles in stellar nucleosynthesis.